Quantum chemistry · Unified chapter

QC gaps: why they matter, H_2 as a minimal diagnostic, ethylene exploration in inZOR-ND, and the discrete-search benchmarks

One report for the work thread on **electronic gaps** near quasi-degeneracy: what we built, which mode does what, and how the pieces connect. Updated 2026-04-03.

Single page · raw H_2 data on this path · figures from inZOR-ND validation exports

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WHAT PROBLEM WE ADDRESS

Near **quasi-degeneracy** (avoided crossings, stretched bonds, twisted π systems), **multi-reference** methods depend critically on the **active space**. A poor choice does not only shift total energies: it can distort **excitation gaps** and state orderings long before a full potential-energy surface is computed. This chapter documents two complementary lines of work:

- **Minimal diagnostic (H_2):** a seven-point H–H scan comparing SA-CASSCF(2,2) **HOMO–LUMO gaps** when orbitals are chosen by **inZOR-**

ND versus a **NOON**-style rule. The goal is a cheap, interpretable signal: does the gap curve behave physically along the scan?

- **Exploratory system (ethylene):** in **inZOR-ND**, a dedicated scene (`scene_ethylene_quasidegen`) feeds **SA-CASSCF(2,2)** gap-related observables back into an **evolving population** (2D/3D, multiple seeds, score variants). The goal is not to replace benchmark solvers but to study **adaptive exploration** under the same chemical stress (near-degeneracy) that motivates strict QC validation elsewhere.

Large-scale **inZOR-ND** results for automatic CAS selection (N_2, Cr_2, eight-benchmark comparative, including ethylene SA-CASSCF) remain the **authoritative discrete-search validation**. This page makes the **scope boundary** explicit so readers never confuse **continuous population exploration** logs with **discrete orbital-set search** benchmark tables — both ship under the inZOR-ND name, but the algorithms and fitness landscapes differ.

Two modes of inZOR-ND on this page

Discrete search (benchmark track)

Bio-adaptive **discrete search** over candidate active spaces (orbital index sets). Used for the **H_2 gap table** (fixed protocol per method) and for all **published CAS benchmarks** linked below.

Why it appears in an H_2 "gap" note: same orbital-choice question as in large molecules, but in the smallest system where the gap trace is easy to read.

Continuous exploration (ethylene scene)

2D/3D **artificial-life** environment: genomes propose moves; a **quantum-chemistry probe** (here SA-CASSCF(2,2)) scores proposals. Used for **ethylene quasi-degeneracy** exploration and validation batches (`legacy` / `local_coherent` / `dwell_coherent` , optional ND).

Why ethylene: richer geometry than H_2 while still highlighting **gap-driven** fitness near problematic electronic regimes.

> **Claim hygiene.** Do not cite **population exploration** logs as **discrete-search benchmark** outcomes, or vice versa. The scientific motif (gaps under quasi-degeneracy) is shared; the algorithms and fitness landscapes are not.

H_2 bond stretch — SA-CASSCF(2,2) HOMO–LUMO gap

Why H_2

Two electrons, minimal active space: the scan isolates **orbital-choice effects** on a single gap observable without conflating with conformational degrees of freedom. If a heuristic mis-picks the active pair at an intermediate bond length, the gap can **fail basic monotonicity** even when energies might still look plausible at isolated points.

Setup

Basis: cc-pVDZ. **Scan:** seven H–H distances from 0.7 Å to 2.5 Å. **Methods compared:** active MO indices from **inZOR-ND** vs **NOON** (see `h2_stretch_results.json` for exact index lists and flags).

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Geometries

Yes
inZOR gap monotone

No
NOON gap monotone

cc-pVDZ
Basis

Gap table (eV)

R (Å)	inZOR gap (eV)	NOON gap (eV)
0.7	10.743743	10.743759
0.9	7.974340	7.974299
1.1	5.677354	5.677354
1.3	3.887806	3.887805
1.6	2.049057	2.049044
2.0	0.764453	**18.143451**
2.5	0.188814	0.188821

NOON at R = 2.0 Å. The gap explodes to ~18.14 eV in this snapshot while neighbouring points are ~2 eV — `monotonic: false` in JSON. That is exactly the kind of **early pathology** a gap diagnostic is meant to catch before trusting longer PECs or dynamics on the same orbital logic.

Interpretation. For this documented run, inZOR-ND's selected pair yields a **smoothly decaying** gap across dissociation; the NOON-tracked pair does not. This does not by itself prove global correctness on all systems; it **motivates** the larger inZOR-ND benchmark programme where energies and multi-geometry stability are scored directly.

Ethylene quasi-degeneracy — inZOR-ND scene

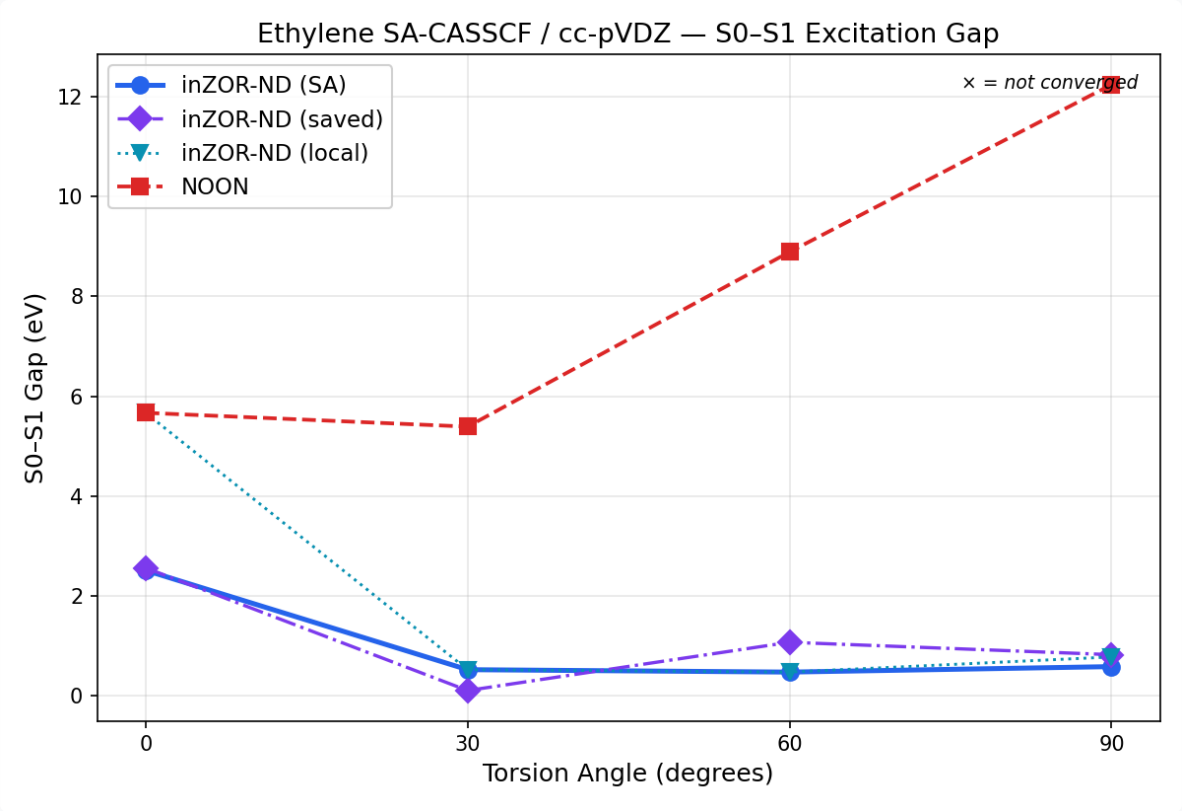
Ethylene 3D (regional validation): published as its **own** entry at the top of the Research Tests page — Ethylene (3D) — Quasi-Degenerate Gap Structure via inZOR-ND. Technical report: ethylene_3d_zor.html.

What we built and why

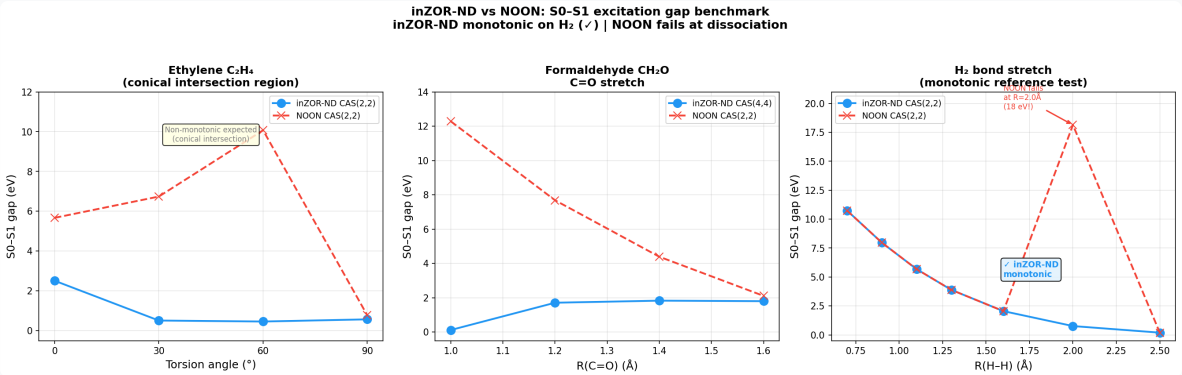
Ethylene near torsional quasi-degeneracy is a classic setting where **single-reference pictures weaken** and SA-CASSCF(2,2) is a minimal multi-reference anchor. In **inZOR-ND** we attach that computation to an **evolving population** so we can ask: under local sensing only (no hand-written policy), does selection pressure in a 2D/3D world still **track chemically meaningful gap structure** when we change scoring rules or dimensionality?

Protocol (engineering summary)

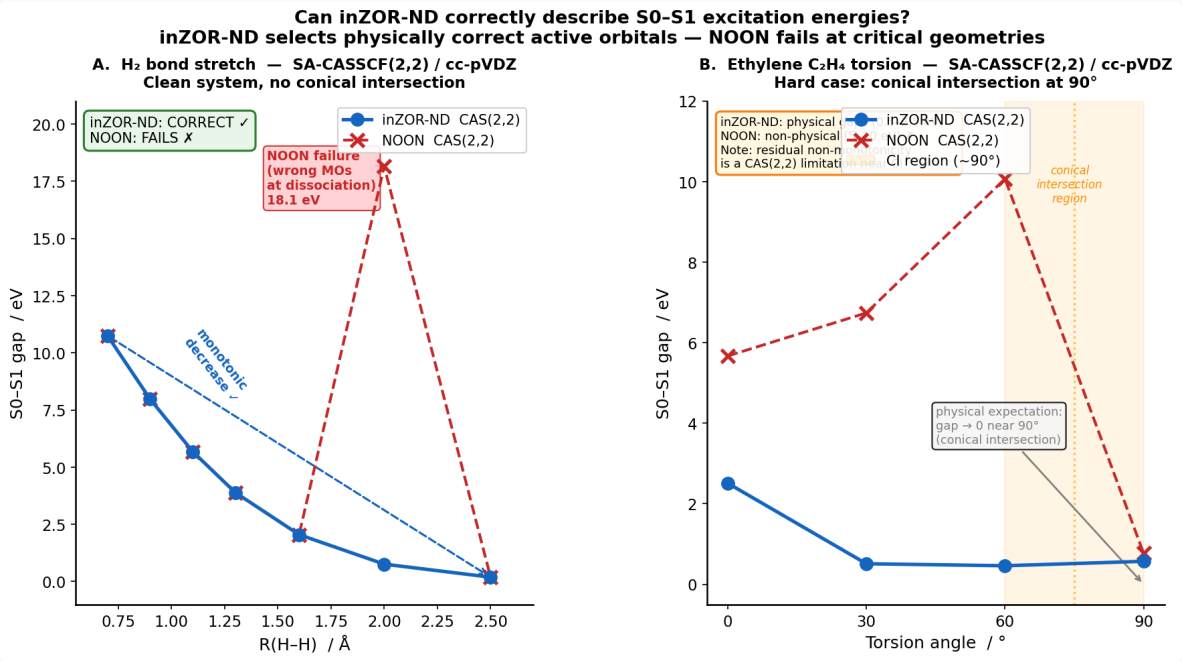
- **Scene path (source tree):** `zor_scenes/scene_ethylene_quasidegen/` — logs, validation scripts, batch outputs as committed there.
- **Electronic model:** SA-CASSCF(2,2); gap-style observables (e.g. S_0–S_1) feed fitness.
- **Validation:** multiple seeds, controls, local grids; **3D** extensions when `ZOR_ETHYLENE_ND` / ND output folders are enabled.
- **Score variants:** `legacy`, `local_coherent`, `dwell_coherent` — to stress-test whether conclusions depend on a particular temporal-smoothing or locality heuristic in the score.



Exported S_0–S_1 gap visual for the ethylene active-space / quasi-degeneracy study (inZOR-ND workspace).



Validation-style comparison across batches or score variants (export from the ethylene validation thread).



Companion panel: metrics / reference answer for the same validation workflow.

Bridge to the inZOR-ND active-space suite

The **comparative eight-benchmark** report includes **ethylene SA-CASSCF** alongside N_2, Cr_2, butadiene, CH_2O, benzene, etc. That is the right place for **discrete-search** head-to-head against NOON-MP2 / AVAS with published energy gaps and advantages.

This unified page adds: (i) a **minimal H_2 gap sanity check** for orbital heuristics, and (ii) an **inZOR-ND continuous-exploration narrative** (ethylene scene) for the same chemical theme. Together they explain

motivation and tooling; the comparative HTML remains the citation target for large-scale CAS discovery claims.

Continue reading

- inZOR-ND — comparative (8 benchmarks)
- N_2 dedicated study
- Cr_2 dedicated study

Reproducibility & PDF

H_2: regenerate from the driver that produced `h2_stretch_results.json` (PySCF / SA-CASSCF stack; pin versions for any formal preprint).

Ethylene (inZOR-ND): reproduce from the source repository — scene assets, `zor_active_space/` exports, and validation logs under `zor_scenes/scene_ethylene_quasidegen/`.

PDF: this HTML is structured as the web companion; a future PDF can lift sections 1:1 with DOI / commit SHA in the front matter.